
IAL MECHANICS 1 · A* EXAMINATION GUIDE

Pearson Edexcel International Advanced Level · Paper WME01/01

Built from 19 question papers and their mark schemes, January 2020 – Summer 2026

How to use this guide. Each topic gives the facts you must own, the method examiners reward, the traps the mark schemes actually penalise, and one real past-paper question solved the way the scheme awards it. Section 2 decodes the marking rules; read it first, because most lost marks in M1 are lost to those rules rather than to the physics.

1. THE PAPER AT A GLANCE

Duration and total	1 hour 30 minutes, 75 marks, 7 or 8 questions (≈ 1.2 minutes per mark)
Constants	$g = 9.8 \text{ m s}^{-2}$; numerical answers after using g to 2 or 3 significant figures
Shape of the paper	Short questions (6–8 marks) first, usually collisions or moments; the final question is almost always a connected-particle pulley problem worth 9–18 marks

Topic frequency across the 19 papers analysed

TOPIC	PAPERS	MARKS	USUAL FORM
Momentum and impulse	19/19	6–8	Direct collision: find a speed, a mass ratio k or an impulse; sometimes combined with friction or a bounce
Moments	19/19	6–12	Rod on two supports or two ropes; “on the point of tilting”; non-uniform rod with unknown centre of mass
Friction	19/19	8–12	Limiting equilibrium; force at an angle; rough slope; “will it stay at rest?”
Vectors in $\mathbf{i}, \mathbf{j}$	19/19	7–14	Resultant forces with $\mathbf{F} = m\mathbf{a}$; ships with position vectors and bearings
Motion graphs	19/19	2–14	Sketch with key values; area = distance, gradient = acceleration; catch-up problems
Inclined planes	17/19	8–14	Resolve along and perpendicular to the slope
Pulley systems	15/19	9–18	Final question; tension, acceleration, force on pulley, second phase after a string goes slack
Vertical motion under gravity	14/19	6–11	Projected up from a height; time, speed, total distance, speed–time sketch
Closest approach, collision, “within d ”	9/19	4–7	Minimise $ \vec{AB} ^2$, or solve $ \vec{AB} ^2 \leq d^2$
“State how you used ...”	9/19	1	One-mark modelling statement (see Section 5)
Lifts, towing, couplings	7/19	6–10	Whole system first, then one body

2. HOW THE MARK SCHEME AWARDS MARKS

M1	Method mark. Typical wording: “correct number of terms, dimensionally correct, condone sign errors and sin/cos confusion”. An equation with every force present but one sign wrong still earns M1.
A1	Accuracy mark, usually for a <i>correct unsimplified equation</i> , then for the answer. Always write the equation in full before simplifying it.
DM1	Method mark that depends on earlier M marks (typically “solve the two equations”). Lose the first M and this falls too.
B1	Independent fact, e.g. $F = \mu R$ seen anywhere (even on the diagram), or $R = mg \cos \alpha$.
A1*	Given answer (“show that”). Needs completely correct working, including the exact trig values used.
ft, cso	Follow-through on your earlier value; “correct solution only” (no slips anywhere).

RULINGS THAT RECUR ACROSS THE 19 SCHEMES

- **g in an impulse or momentum equation is M0** (dimensionally wrong). In $F = ma$, a missing m or missing g is M0 or an A error.
- **After using $g = 9.8$, give a decimal to 2 or 3 s.f.** Final answers such as $5/8$, $32/125$, $10.5u/3$ or “ $2g - 3$ ” were marked A0. Over-accuracy is penalised once per question; $g = 9.81$ is penalised once per question.
- Symbolic answers in m and g (e.g. $\frac{15}{4}mg$) are expected when the question asks “in terms of”.
- **One suvat equation across a change of acceleration is M0.** Assuming $u = 0$ when it is not stated is M0.
- Moments: **M0 if the tilting reaction is never set to zero**; M0 if the two reactions are assumed equal.
- An inequality ($P \leq \dots$) that never becomes an equation loses the final accuracy marks.
- Graph sketches: **a solid vertical line is B0**; label start, end and turning values.
- Modelling statements: **any incorrect extra statement is B0**; for a pulley, “pulley not mentioned” is B0.
- Vectors: column vectors are accepted in working, but give the final answer in **i, j** form when asked.

3. KINEMATICS IN A STRAIGHT LINE

FACTS TO OWN

suvat (constant acceleration only): $v = u + at$, $s = ut + \frac{1}{2}at^2$, $s = vt - \frac{1}{2}at^2$, $v^2 = u^2 + 2as$, $s = \frac{1}{2}(u + v)t$.

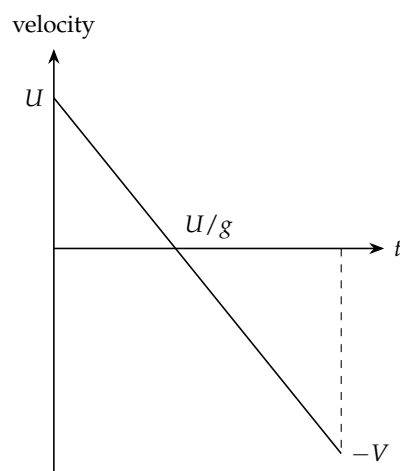
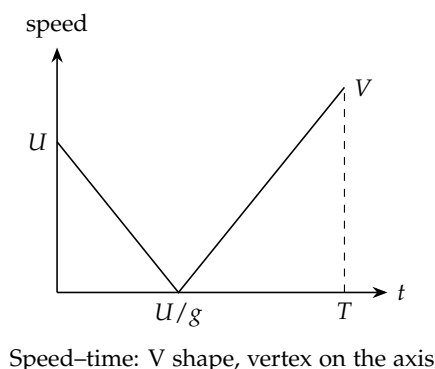
Vertical motion: take up as positive, so $a = -g$. Time to the top = U/g ; greatest height = $U^2/2g$. A point h below the start has displacement $s = -h$.

Graphs: on a velocity–time graph, gradient = acceleration and area = displacement; on an acceleration–time graph, area = change in velocity; on a displacement–time graph, gradient = velocity.

Distance in the n th second = $s(n) - s(n - 1)$. **Total distance** for up-and-down motion = distance up + distance down (not the displacement).

Method

1. Draw a line diagram; mark the positive direction and every known value of s, u, v, a, t .
2. Split the motion into stages wherever the acceleration changes. The final v of one stage is the u of the next.
3. For a time with s given, $s = ut + \frac{1}{2}at^2$ gives a quadratic: keep both roots and decide which is meant. "Time for which the particle is above C " is the *difference* of the two roots.
4. Catch-up problems: write each distance as a function of the same clock time t (a vehicle starting 10 s later has been moving for $t - 10$ seconds), then equate.



EXAMINER TRAPS

- Speed-time sketches must reach the time axis at the top of the flight and have equal-magnitude gradients; the second line is longer when the particle lands below its start.
- The speed at impact must be given as a positive number even if your velocity is negative.
- Do not stop the dashed line or the graph with a solid vertical line at the end (B0).

Worked example — Summer 2025, Q3 (8 marks). Projected vertically upwards with speed U from A , 12 m above the ground at B ; it reaches B after 4 s. Find U , the speed at B , and sketch the speed-time graph.

WORKING	MARK
(a) Up positive, $s = -12$, $t = 4$, $a = -9.8$: $-12 = 4U - \frac{1}{2}(9.8)(4^2)$	M1 A1
$4U = 66.4 \Rightarrow U = 16.6$ ("2g - 3" left as the answer scored A0)	A1
(b) $v = u + at = 16.6 - 9.8(4) = -22.6$, so speed = 22.6 m s^{-1}	M1 A1 A1
(c) V shape, vertex on the t -axis at $t = 1.69$, second line longer; values 16.6, 4 and 22.6 marked	B1 B1ft

Worked example — January 2024, Q6(c) (4 marks). Projected upwards at 24 m s^{-1} from A; C is 10 m above A. Find the time for which the particle is above C.

WORKING	MARK
$10 = 24t - 4.9t^2 \Rightarrow 4.9t^2 - 24t + 10 = 0$	M1 A1
$t = \frac{24 \pm \sqrt{380}}{9.8}$, so $t_1 = 0.460$, $t_2 = 4.438$; time above C = $t_2 - t_1$	M1
$= 3.98 \text{ s}$	A1

Worked example — October 2020, Q8 (8 marks). An acceleration–time graph: 3 m s^{-2} for $0 \leq t \leq 4$ (from rest), 1.5 m s^{-2} for $4 \leq t \leq 24$, -4 m s^{-2} for $24 \leq t \leq 26$. Find the distances.

WORKING	MARK
(a) First 4 s: $s = \frac{1}{2}(3)(4^2) = 24 \text{ m}$	M1 A1
(b) Velocities from areas under the a – t graph: $v(4) = 12$, $v(24) = 12 + 1.5(20) = 42$, $v(26) = 42 - 8 = 34$	B1
$4 \leq t \leq 24$: trapezium $\frac{1}{2}(12 + 42)(20) = 540$	M1 A1ft
$24 \leq t \leq 26$: trapezium $\frac{1}{2}(42 + 34)(2) = 76$	M1 A1ft
Total = $24 + 540 + 76 = 640 \text{ m}$	A1

4. FORCES, FRICTION AND EQUILIBRIUM OF A PARTICLE

FACTS TO OWN

Friction: $F \leq \mu R$. “On the point of moving” or “limiting” means $F = \mu R$. Friction opposes the motion, or the motion that would happen without it.

Normal reaction always comes from resolving perpendicular to the surface *with every force included*:

- pull P at θ above a horizontal surface: $R = mg - P \sin \theta$; push at θ below: $R = mg + P \sin \theta$;
- horizontal force X on a slope of angle α : $R = mg \cos \alpha + X \sin \alpha$, and along the slope $X \cos \alpha$ is the useful component;
- a ring on a vertical rod: R is horizontal and friction acts vertically, up or down.

Rest on a rough slope with no other force $\iff \mu mg \cos \alpha \geq mg \sin \alpha \iff \mu \geq \tan \alpha$.

Exact trig used by the papers: $\tan \alpha = \frac{3}{4} \Rightarrow \sin \alpha = \frac{3}{5}, \cos \alpha = \frac{4}{5}$; $\tan \alpha = \frac{5}{12} \Rightarrow \frac{5}{13}, \frac{12}{13}$; $\tan \alpha = \frac{7}{24} \Rightarrow \frac{7}{25}, \frac{24}{25}$. Use these fractions, never a rounded angle.

Method

1. Draw a large force diagram: weight, R , F (with its direction for *this* case), and every applied force with its angle.
2. Resolve perpendicular to the surface first to find R . Then resolve parallel.
3. “Greatest” and “least” force questions are two separate limiting cases: friction points one way for impending motion up, the other way for impending motion down.
4. Three forces in equilibrium may also be solved with a triangle of forces or the sine rule; resolving in two directions is always accepted.

Worked example — October 2020, Q3 (8 marks). A 10 kg particle on a rough slope with $\tan \alpha = \frac{3}{4}$, $\mu = 0.5$, held by a force P up the slope. Find R , the greatest P and the least P .

WORKING	MARK
(a) $R = 10g \cos \alpha = 8g = 78.4 \text{ N}$	M1 A1
(b) Greatest P : about to move up, friction acts down the slope; $F = 0.5R$ $P = 10g \sin \alpha + F = 6g + 4g$ $P = 98 \text{ N}$	B1 M1 A1 A1
(c) Least P : about to slip down, friction acts up the slope: $P = 10g \sin \alpha - F = 2g = 19.6 \text{ N}$	M1 A1

EXAMINER TRAPS

- Using the old value of R after a force has been removed or changed: the scheme gives M0 for “the previous R ”.
- “Determine whether it remains at rest”: find the maximum friction μR , compare it with the component of weight down the slope, and *write the conclusion in words*.
- “Describe what happens” (e.g. $P = 14$): compare the driving component with μR ; if it is less, friction equals the driving component, not μR .

5. NEWTON’S LAWS AND CONNECTED PARTICLES

FACTS TO OWN

Newton’s second law: resultant force in the direction of the acceleration $= ma$.

Lift: for a person of mass m in a lift accelerating upward at a , $R - mg = ma$. For the whole lift, use the total mass with the cable tension.

Towing and couplings: one equation for the whole system (no tension) and one for a single body. A coupling modelled as a light rod can carry a *thrust*.

Pulleys: one equation for each particle, each with its *own* mass in ma ; same T and same $|a|$ throughout.

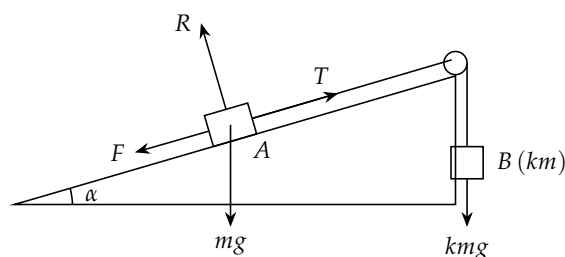
Force on a pulley $= 2T \cos(\frac{1}{2}\phi)$, where ϕ is the angle between the two string directions; it acts along the bisector of that angle.

- both strings vertical: $2T$; table edge (one horizontal, one vertical): $\sqrt{2}T$ at 45° below the horizontal;
- slope of angle α with the other particle hanging: $\phi = 90^\circ - \alpha$.

Second phase: when a hanging particle hits the ground or a string breaks, the string goes slack. The other particle keeps its speed at that instant and now has a *new* acceleration: $-\mu g$ on a rough table; $-g(\sin \alpha + \mu \cos \alpha)$ moving up a rough slope; $-g$ if moving freely upwards.

"STATE HOW YOU USED ..."	ANSWER THAT SCORED B1 (GIVE ONE STATEMENT ONLY)
Pulley is smooth	The tension is the same on both sides of the pulley
String is light	The tension is the same throughout each straight section of the string
String is inextensible	Both particles have the same magnitude of acceleration
Beam modelled as a rod	The beam is rigid / remains straight / does not bend
Rod is uniform	Its weight acts at the midpoint
Block modelled as a particle	Its mass (weight) is concentrated at a single point
Improve the model	e.g. the pulley is not smooth, the string is not light, the particles have size

Worked example — October 2023, Q7 (13 marks). Block A (mass m) on a rough slope, $\tan \alpha = \frac{7}{24}$, $\mu = \frac{1}{3}$, joined over a smooth pulley at the top to B (mass km) hanging. A moves up the slope and $T = \frac{4}{3}mg$. Find a , k and the force on the pulley.



WORKING	MARK
$\sin \alpha = \frac{7}{25}$, $\cos \alpha = \frac{24}{25}$; perpendicular: $R = mg \cos \alpha = \frac{24}{25}mg$	M1 A1
$F = \frac{1}{3}R = \frac{8}{25}mg$	M1
Along the slope for A: $\frac{4}{3}mg - mg \sin \alpha - F = ma$	M1 A1
$a = \frac{4}{3}g - \frac{7}{25}g - \frac{8}{25}g = \frac{11}{15}g$	A1
For B: $kmg - \frac{4}{3}mg = km \cdot \frac{11}{15}g \Rightarrow \frac{4}{15}k = \frac{4}{3} \Rightarrow k = 5$	M1 A1 A1
Angle between strings = $90^\circ - \alpha$; force = $2T \cos\left(\frac{90^\circ - \alpha}{2}\right)$	M1 A1
$\cos\left(\frac{90^\circ - \alpha}{2}\right) = \sqrt{\frac{1 + \sin \alpha}{2}} = \frac{4}{5}$, so force = $2 \cdot \frac{4}{3}mg \cdot \frac{4}{5} = \frac{32}{15}mg$	dM1 A1

Worked example — January 2020, Q7(e) (5 marks) — the second-phase pattern. A (4 kg, rough table, $\mu = 0.625$) starts 2 m from the pulley; B falls 1.4 m from rest with $a = 0.7$ and hits the floor. Does A reach the pulley?

WORKING	MARK
Speed when B lands: $v^2 = 2(0.7)(1.4) = 1.96$, $v = 1.4 \text{ m s}^{-1}$	M1
String slack; for A: $-\mu(4g) = 4a' \Rightarrow a' = -\frac{5}{8}g$	M1
$0 = 1.96 - 2\left(\frac{5}{8}g\right)s \Rightarrow s = 0.16 \text{ m}$	M1 A1
$1.4 + 0.16 = 1.56 < 2$, so A does not reach the pulley (comparison with 2 must be seen)	A1 cso

Worked example — Summer 2025, Q6 (11 marks) — particle projected up a rough slope. $\tan \theta = \frac{3}{4}$, $\mu = \frac{1}{4}$. Show the deceleration is $\frac{4}{5}g$; find U if it stops after 1.5 m; find the least X up the slope to hold a 2 kg package there.

WORKING	MARK
$R = mg \cos \theta = \frac{4}{5}mg$; $F = \frac{1}{4}R = \frac{1}{5}mg$	M1 M1 A1
$F + mg \sin \theta = ma \Rightarrow a = \frac{1}{5}g + \frac{3}{5}g = \frac{4}{5}g$ (“0.8g” as the final line scored A0 in this show-that)	M1 A1 A1*
$0 = U^2 - 2 \cdot \frac{4}{5}g \cdot 1.5 \Rightarrow U = 4.85 \text{ m s}^{-1}$	M1 A1
Least X : about to slip down, friction up the slope: $X + \frac{1}{4}(2g)\frac{4}{5} = 2g \cdot \frac{3}{5} \Rightarrow X = 0.8g = 7.84 \text{ N}$	M1 A1 A1

EXAMINER TRAPS

- Writing $T = 3mg$ (the weight) for a hanging particle that is accelerating loses the dependent marks.
- The mass in the ma term decides which body an equation is for; the whole-system equation cannot give T .
- In a “show that”, the last line must match the printed form exactly ($\frac{4}{5}g$, not 0.8g).
- Converting tonnes: 15 tonnes = 15 000 kg.

6. MOMENTUM AND IMPULSE

FACTS TO OWN

Conservation of linear momentum: $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$, with every velocity signed relative to one chosen positive direction.

Impulse = $m(v - u)$, units N s. The impulse on P from Q is equal and opposite to the impulse on Q from P , so either particle’s change in momentum gives the impulse.

Coalescing / string becoming taut: both move with one common velocity afterwards.

Bounce on the ground: directions reverse, so the impulse magnitude is $m(v + u)$ in terms of speeds.

Method

1. Draw “before” and “after” diagrams with masses and arrows; mark the positive direction.
2. If a direction after the collision is unknown, draw it in the positive direction; a negative answer means the opposite way, and you must state the direction when asked.
3. If only a *speed* is given and the direction is not stated, there are two cases: this is where “find the two possible values of k ” comes from (Summer 2021, Q1: $k = \frac{28}{3}$ or $\frac{22}{3}$; October 2025, Q2).
4. “Direction reversed” gives an inequality on a velocity, which is how “show that $m_2 > 4m_1$ ” type parts are done.

Worked example — Summer 2025, Q1 (6 marks). *P* (mass km , speed $5u$) catches *Q* (mass $3m$, speed $2u$) in the same direction. Afterwards *P* moves at $2u$ in the same direction; the impulse on *Q* has magnitude $4.5mu$. Find v and k .

WORKING	MARK
(a) Impulse on <i>Q</i> : $4.5mu = 3m(v - 2u)$ $v = 3.5u$ (“ $10.5u/3$ ” as the final answer was A0)	M1 A1 A1
(b) Impulse on <i>P</i> is $-4.5mu$: $-4.5mu = km(2u - 5u)$ $k = 1.5$	M1 A1 A1

EXAMINER TRAPS

- Any g in the momentum or impulse equation: M0.
- Impulse magnitude must be positive; state units (N s) when the question says “stating the units”.
- Impulse after a frictional slide (January 2025, Q3; October 2021, Q2): the speed just before the collision comes from the friction stage, not from the original projection.

7. MOMENTS

FACTS TO OWN

Equilibrium of a rod: (i) resultant vertical force is zero; (ii) the sum of moments about *any* point is zero. Moment = force $\times$ perpendicular distance.

On the point of tilting about C: the reaction (or tension) at the *other* support is zero. Write this down explicitly.

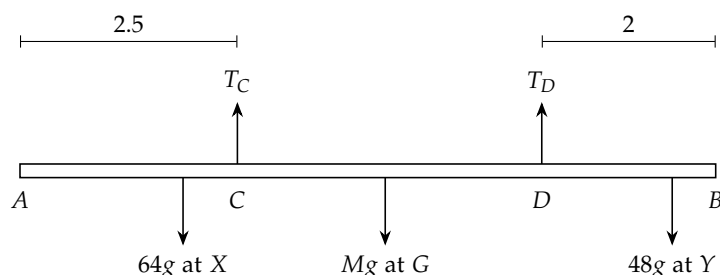
Uniform rod: weight at the midpoint. **Non-uniform rod:** weight at an unknown distance (d or x) from a named end.

Ranges: “range of possible values of M ” means each reaction ≥ 0 (and any string tension $\leq$ its breaking value).

Method

1. Draw the rod with all distances measured from one end; mark every force.
2. Take moments about the point where the force you do not want acts, usually a support.
3. Two scenarios (two different loads, two tilting positions) give two equations in the two unknowns (M and d , or W and x). Eliminate.
4. Resolving vertically is the quickest second equation when both reactions are involved.

Worked example — October 2020, Q4 (8 marks). Non-uniform beam AB , 8 m, mass M , centre of mass d from A ; ropes at C ($AC = 2.5$) and D ($DB = 2$). A 64 kg gymnast at X ($AX = 1.875$): on the point of tilting about C . A 48 kg gymnast at Y ($YB = 0.5$): on the point of tilting about D . Find M .



WORKING	MARK
Gymnast at X , tilting about C : $T_D = 0$	M1
$M(C)$: $64g(0.625) = Mg(d - 2.5) \Rightarrow M(d - 2.5) = 40$	M1 A1
Gymnast at Y , tilting about D : $T_C = 0$	M1
$M(D)$: $48g(1.5) = Mg(6 - d) \Rightarrow M(6 - d) = 72$	M1 A1
Add: $3.5M = 112 \Rightarrow M = 32$	DM1 A1

EXAMINER TRAPS

- Using a reaction value from one scenario in the other scenario.
- Measuring the distance of the centre of mass from the wrong end; state “ d from A ” on the diagram.
- Rod on a ledge or step: the only possible pivot is the edge of the ledge, and contact elsewhere is zero at tilting.

8. VECTORS: FORCES AND MOTION IN TWO DIMENSIONS

FACTS TO OWN

$|a\mathbf{i} + b\mathbf{j}| = \sqrt{a^2 + b^2}$. Resultant $= \sum \mathbf{F}$; $\sum \mathbf{F} = m\mathbf{a}$; equilibrium $\Rightarrow \sum \mathbf{F} = \mathbf{0}$.

Constant velocity: $\mathbf{r} = \mathbf{r}_0 + \mathbf{v}t$, and $\mathbf{v} = \frac{\mathbf{r}_2 - \mathbf{r}_1}{t_2 - t_1}$. **Constant acceleration:** $\mathbf{v} = \mathbf{u} + \mathbf{a}t$.

Relative position: $\overrightarrow{AB} = \mathbf{b} - \mathbf{a}$. The bearing of B from A is the direction of $\overrightarrow{AB}$, measured clockwise from north ($\mathbf{j}$), in three figures.

WORDING IN THE QUESTION	TRANSLATE TO
Parallel to / in the direction of $p\mathbf{i} + q\mathbf{j}$	Components in the ratio $p : q$ with the same signs; opposite direction means $\lambda < 0$
B is due north / due east of A	$\mathbf{i}$ -component / $\mathbf{j}$ -component of $\vec{AB}$ is zero (the other positive)
B is north-east / north-west of A	$\vec{AB} = k(\mathbf{i} + \mathbf{j}) / k(-\mathbf{i} + \mathbf{j})$ with $k > 0$
They collide	$\mathbf{r}_A = \mathbf{r}_B$: both components give the <i>same</i> t
Show they never collide	The t that zeroes one component of $\vec{AB}$ does not zero the other
Closest approach	Minimise $ \vec{AB} ^2$: complete the square or differentiate
Within d km / visible	$ \vec{AB} ^2 \leq d^2$; time = difference of the roots
Same speed / same direction	$ \mathbf{v}_A ^2 = \mathbf{v}_B ^2$ / components in the same ratio
Clock times and units	7:36 am is 0.6 h after 7 am; $\text{km h}^{-1} \div 3.6 = \text{m s}^{-1}$

Worked example — January 2022, Q8(c)(d) (8 marks). $\vec{AB} = [(12t - 10)\mathbf{i} + (24t - 10)\mathbf{j}]$ km. Find the shortest distance between the ships and the bearing of B from A at that moment.

WORKING	MARK
$ \vec{AB} ^2 = (12t - 10)^2 + (24t - 10)^2 = 720t^2 - 720t + 200$	M1
$= 720 \left(t - \frac{1}{2}\right)^2 + 20$ (or differentiate: $1440t - 720 = 0$)	M1 A1
Minimum at $t = \frac{1}{2}$	DM1
$\vec{AB} = -4\mathbf{i} + 2\mathbf{j}$, so distance $= \sqrt{20} = 4.47$ km	M1 A1
B is $\tan^{-1} \frac{4}{2} = 63.4^\circ$ west of north: bearing $= 360^\circ - 63.4^\circ = 297^\circ$	M1 A1

EXAMINER TRAPS

- $\vec{AB} = \mathbf{a} - \mathbf{b}$ (wrong way round) loses the A mark in a given-answer part.
- Starting a collision question with $\mathbf{r}_A = \mathbf{r}_B$ and solving only one component is M0; check both.
- Always sketch the vector before quoting an angle or bearing; most bearing errors are quadrant errors.
- “Solutions relying on calculator technology are not acceptable” (Summer 2026): show the algebra, e.g. completing the square, in full.

9. RECENT TRENDS, 2024 – 2026

The core seven question types have not changed, but recent papers add a twist to familiar set-ups. Expect at least two of these.

PAPER	NEW TWIST
Summer 2026	Displacement–time graph with an impulse at the turning point; ball whose speed halves at each bounce (time above 4 m); boat visible within 9 km (quadratic inequality); direction of the force on a pulley in terms of α
January 2026	Blocks of 0.2 kg added one at a time until motion starts: find the integer n from an inequality, then the initial acceleration; particle sliding through A, B, C on a rough surface
October 2025	Resultant of magnitude $10\sqrt{2}$ giving two possible forces; motorcycle catching a car read off a velocity–time graph; two particles projected from different heights at different times
Summer 2025	Two boats with vector accelerations: same speed (exact time) and same direction (quadratic); rebound impulse at the ground followed by a slack-string phase
January 2025	Impulse $\rightarrow$ frictional slide $\rightarrow$ collision in one question; “Determine whether P remains at rest” after a string breaks
October 2024	Impulse from a bounce using rebound height; string breaks mid-motion; ball B projected 1 s after ball A

10. THE A* CHECKLIST

BEFORE THE EXAM

- Work every one of the 19 papers under timed conditions, then mark against the scheme line by line, not just the final answers.
- Know the exact trig triples (3-4-5, 5-12-13, 7-24-25) and the force-on-pulley rule by heart.
- Memorise the seven modelling statements in Section 5 word for word.

IN THE EXAM ROOM

1. **Diagram first**, with the positive direction and every force or velocity labelled.
2. **Write each equation in full and unsimplified** before tidying: that line is usually an A1.
3. Name what you are doing: “Resolve $\perp$ plane”, “ $M(C)$ ”, “CLM”, “For B : $F = ma$ ”.
4. Keep exact values (fractions of g , surds) until the final line; then give 3 s.f. after any use of $g = 9.8$.
5. Check “in terms of” requirements, units (Ns for impulse), positive speeds, and three-figure bearings.
6. For a “show that”, finish on exactly the printed form.
7. Budget about 1.2 minutes per mark; if stuck, write the equations you can, since method marks do not need the final answer.